```c
#include <stdio.h>

void addition(int a[10][10], int b[10][10], int r, int c)
{
    int i, j;

    printf("\nMatrix Addition:\n");

    for(i = 0; i < r; i++)
    {
        for(j = 0; j < c; j++)
        {
            printf("%d\t", a[i][j] + b[i][j]);
        }
        printf("\n");
    }
}

void multiplication(int a[10][10], int b[10][10]
```

```
void multiplication(int a[10][10], int b[10][10],
                    int r1, int c1, int r2, int c2)
{
    int i, j, k;
    int result[10][10];

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            result[i][j] = 0;

            for(k = 0; k < c1; k++)
            {
                result[i][j] =
                result[i][j] + a[i][k] * b[k][j];
            }
        }
    }

    printf("\nMatrix Multiplication:\n");
```

```
    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c2; j++)
        {
            printf("%d\t", result[i][j]);
        }
        printf("\n");
    }
}

void transpose(int a[10][10], int r, int c)
{
    int i, j;

    printf("\nTranspose:\n");

    for(i = 0; i < c; i++)
    {
        for(j = 0; j < r; j++)
        {
            printf("%d\t", a[j][i]);
        }
```

```
}


nt main()

    int a[10][10], b[10][10];
    int r1, c1, r2, c2;
    int i, j;

    /* Read Matrix A */
    printf("Enter rows and columns of Matrix A: ");
    scanf("%d %d", &r1, &c1);

    printf("Enter elements of Matrix A:\n");

    for(i = 0; i < r1; i++)
    {
        for(j = 0; j < c1; j++)
        {
            scanf("%d", &a[i][j]);
        }
```

```
printf("\nEnter rows and columns of Matrix B: ");
scanf("%d %d", &r2, &c2);

printf("Enter elements of Matrix B:\n");

for(i = 0; i < r2; i++)
{
    for(j = 0; j < c2; j++)
    {
        scanf("%d", &b[i][j]);
    }
}

/* Addition */
if(r1 == r2 && c1 == c2)
{
    addition(a, b, r1, c1);
}
else
{
    printf("\nMatrix Addition is not possible.\n");
}
```

```
    if(c1 == r2)
    {
        multiplication(a, b, r1, c1, r2, c2);
    }
    else
    {
        printf("\nMatrix Multiplication is not possible.\n");
    }

    /* Transpose of Matrix A */
    transpose(a, r1, c1);

    return 0;
}
```